

ASSIGNMENT #4: GEOSPATIAL MAPPING AND FIELD VARIABILITY VISUALIZATION

01. **Task:** Plot your field boundaries as the base layer.
02. **Task:** Overlay a second spatial dataset.
 - a. *Example:* Color the fields by their dominant **Soil Type** or overlay the locations of the nearest **Weather Stations**.
03. **Task:** Add a legend, a north arrow (if applicable), and clear labels.

Step 4: Interpret Field Variability

Below your map in the notebook, create a Markdown cell titled "Interpretation."

01. Write a brief paragraph (3-5 sentences) explaining what the map shows.
02. *Example:* "The map reveals that the western cluster of fields is predominantly comprised of silty clay loam, which correlates with the drainage issues noted in the initial acquisition report."

Step 5: Design Your Dashboard Map

01. Refine your map for a "Dashboard View." This should be your most polished spatial visualization yet.
02. **Task:** Save the final map as `output/dashboard_assets/field_spatial_map.png`.

Step 6: Update Your Project Tracker

Open `docs/project/assignment-01-tracker.md`.

- Log the spatial datasets you combined.
- Note any challenges with data alignment or CRS reprojection.

GRADING CRITERIA (8 POINTS TOTAL)

CRITERION	POINTS	DESCRIPTION
SPATIAL INTEGRATION	3	Successfully combined and aligned at least two distinct spatial datasets.
INTERPRETATION	2	Provided a clear, logical explanation of what the spatial patterns indicate.
DASHBOARD DESIGN	2	Created 1 polished map visualization ready for the final dashboard.
TECHNICAL EXECUTION	1	Correct use of GeoPandas and appropriate Coordinate Reference Systems.

ASSIGNMENT OVERVIEW

FIELD	VALUE
ASSIGNMENT NUMBER	4
TITLE	Geospatial Mapping and Field Variability Visualization
GRADE VALUE	8 points
DUE DATE	March 12, 2026, 11:59 PM PT

OBJECTIVES

After completing this assignment, you will be able to:

- Create geospatial visualizations combining at least two different spatial datasets.
- Master the use of `GeoPandas` for layering boundaries and environmental data.
- Understand the impact of `spatial variability` (soil, topography, or location) on field performance.
- Design high-quality maps that communicate complex agricultural insights at a glance.

INSTRUCTIONS

Step 1: Create Your Feature Branch

Open your terminal in VS Code and switch to a new branch for your mapping work:

```
git checkout -b feature/assignment-04-mapping
```

Step 2: Set Up Your Geospatial Notebook

01. **Create a new notebook:** `notebooks/04_field_mapping.ipynb`.
02. **Load your spatial data:** Import your field boundaries (GeoJSON) and your NRCS soil data.
03. **Check CRS:** Verify that both layers use the same Coordinate Reference System (e.g., EPSG:4326). If not, use `.to_crs()` to align them.

Step 3: Create an Integrated Map (The "2 Layer" Task)

You must create a visualization that shows how different datasets overlap.